



IPSEC REGULATORY AUDIT & CRYPTOGRAPHIC HEALTH

IPsec VPN Security Assessment & Framework Audit

Comprehensive evaluation against NIST SP 800-77 Rev. 1, RFC 8221, and NSA CNSA 2.0 cryptographic mandates

35

GRADE F · CRITICAL
NON-COMPLIANCE

CONDITIONAL COMPLIANCE · ATTENTION REQUIRED

The IPsec deployment scored 35.0 out of 100 (GRADE F · CRITICAL NON-COMPLIANCE), indicating robust transport confidentiality via AES-256-GCM / 3DES (Heterogeneous) encryption. However, critical gaps in Child SA forward secrecy and static shared key authentication require remediation under NIST SP 800-77 Rev 1.

Section 1: Scope & Regulatory Framework

Definition of audit benchmarks (NIST SP 800-77 Rev. 1, RFC 8221, NSA CNSA 2.0)

AUDIT SCOPE & TARGET GATEWAYS

Multi-Link Mesh Deployment (3 Tunnels across Tactical ...)

ASSESSMENT TIMEFRAME

Live Telemetry (FULL): +0.0s - +15.0s (15.0s)

REGULATORY BENCHMARK	TITLE & JURISDICTION	MANDATED CRYPTOGRAPHIC REQUIREMENTS	AUDIT COMPLIANCE VERDICT
NIST SP 800-77 Rev. 1	Guide to IPsec VPNs	Mandates AEAD authenticated ciphers (AES-GCM), enforces Perfect ...	CONDITIONAL GAP
RFC 8221	ESP and AH Cryptographic Requirements	Classifies AES-GCM as MUST implement; strictly deprecates 3DES (...)	COMPLIANT

Section 2: Cryptographic Health Audit

Granular evaluation of symmetric ciphers (AES-GCM vs. CBC/3DES) and hashing integrity (SHA-256 vs. SHA-1/MD5)

SYMMETRIC ENCRYPTION CIPHER

APPROVED

AES-256-GCM / 3DES
(Heterogeneous)

Operating Mode: AEAD (Galois/Counter Mode)
Integrity Tag: 128-bit ICV Authentication Tag
Key Length: 256 bits

AEAD ciphers combine confidentiality and message integrity in a single pass, eliminating padding oracle vectors (POODLE...)

HASHING & MESSAGE INTEGRITY

APPROVED

SHA-256 / SHA-1

Digest Size: 256 bits (SHA-256)
Collision Resistance: Verified (FIPS 180-4 Compliant)
Deprecated Primitives: MD5 / SHA-1 Not Detected

Collision-resistant cryptographic hashing verified. No legacy SHA-1 or MD5 digests detected.



KEY MANAGEMENT, PROTOCOL & STATE INTEGRITY

Key Exchange & Protocol Integrity

Diffie-Hellman modulus size verification, post-quantum posture, sequence rollover safeguards, and anti-replay integrity

Section 3: Key Management & Exchange Audit

Verification of Diffie-Hellman modulus size, post-quantum posture, and PFS rekey enforcement

DIFFIE-HELLMAN MODULUS STRENGTH APPROVED ECP-256 / MODP-1024 256-bit Elliptic Curve (~3072-bit RSA equivalent) Meets NIST SP 800-57 Part 1 Rev 5 discrete logarithm strength recommendations.	PERFECT FORWARD SECRECY (PFS) REKEY CRITICAL NON-COMPL Disabled (Re-uses IKE Master Secret) IKE SA: Active (DH Group 19) -11 pts Absence of ephemeral DH exchange during Child SA rekey means compromise of long-term credentials enables retrospec...	POST-QUANTUM CRYPTOGRAPHIC POSTURE TRANSITION CLASSICAL ECC (Pre-Quantum) Target: Hybrid ML-KEM-768 (Kyber) Migrate to Hybrid ML-KEM-768 (Kyber) + ECP-256 per NSA CNSA 2.0 transition mandate (2026-2030).	SA ROTATION LIFETIMES & LIMITS ALIGNED 3600 seconds (1 Hour) 1,000 MB (1 GB Transfer Threshold) Strict rekey intervals constrain cryptanalytic exposure window under high packet volumes.
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Section 4: Protocol & State Integrity

Anti-replay window sizing, sequence rollover safeguards, and IKE mode security (e.g., IKEv1 Aggressive Mode detection)

ANTI-REPLAY WINDOW SIZING ACTIVE & VERIFIED 64-bit sliding window (RFC 4303) Kernel IPsec state maintains strict packet sequence bitmap. Zero replay packet injection attempts observed.	SEQUENCE ROLLOVER SAFEGUARDS HARDENED 64-bit Extended Sequence Numbers (ESN) Protects against 2^{32} sequence cycling attacks under 10Gbps+ multi-core line-rate traffic. Safe for >500 years.	IKE MODE SECURITY & DOWNGRADE PROTECTED IKEv2 (RFC 7296) IKEv1 Aggressive Mode NOT detected. Unencrypted identity hashing is prevented, neutralizing offline dictionary cracking.
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SIDE-CHANNEL AUDIT & COMPREHENSIVE VULNERABILITY REGISTER

Side-Channel Audit & Threat Matrix

Traffic Flow Confidentiality (TFC) leakage, itemized CVSS vulnerability register, and post-remediation posture

Section 5: Side-Channel & Metadata Vulnerability Audit

Leakage score based on packet shape predictability, burstiness, and missing Traffic Flow Confidentiality (TFC) padding

SIDE-CHANNEL LEAKAGE SCORE

74

MODERATE RISK

PACKET SHAPE PREDICTABILITY

HIGH PREDICTABILITY

Histogram analysis shows distinct clustering around 1420B (video stream) and 180B (audio codec). Clear a...

BURSTINESS & IAT TIMING

MODERATE LEAKAGE

Inter-arrival timing demonstrates regular 20ms frame intervals, leaking interactive VoIP/stream cadences.

TFC PADDING AUDIT

MISSING / DISABLED (-3 pts)

Activate Traffic Flow Confidentiality (TFC) padding per RFC 4303 §2.7 to mask packet size signatures.

Section 6: Comprehensive Threat & Vulnerability Matrix

Detailed register of discovered vulnerabilities, CVSS base/environmental scores, and audit remediation mandates

VULN ID	VULNERABILITY & THREAT VECTOR	CVSS v3.1 BASE / ENV	SEVERITY	REMEDIATION MANDATE
AV-VULN-2026-001	Absence of Child SA Perfect Forward Secrecy	Base: 7.5 Env: 6.8	HIGH	NIST SP 800-77 Rev. 1 §4.2.3
AV-VULN-2026-002	Static Pre-Shared Key (PSK) Mutual Authentication	Base: 6.5 Env: 5.9	MEDIUM	RFC 8221 / Enterprise PKI Standard
AV-VULN-2026-003	Missing Traffic Flow Confidentiality (TFC) Padding	Base: 4.3 Env: 3.8	MEDIUM	RFC 4303 §2.7 Side-Channel Defense
AV-VULN-2026-004	Anti-Replay Window Verification	Base: 0 Env: 0	LOW	RFC 4303 §3.4.3 Anti-Replay

0

CRITICAL

1

HIGH

2

MEDIUM

1

LOW

PROJECTED POST-REMEDIATION POSTURE

CURRENT SCORE	PROJECTED GAIN	PROJECTED SCORE	PROJECTED GRADE
35 / 100	+19 pts	54 / 100	GRADE A+ - FULLY HARDENED

Executing the top two priority directives (enforcing PFS Diffie-Hellman Group 19 on Child SAs and migrating from PSK to X.509 enterprise PKI certificates) immediately eliminates 16 penalty points, elevating the IPsec deployment to complete NIST SP 800-77 Rev. 1 compliance.